



## CONTACT

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## SKILLS

### PERFORMANCE

Profiling, benchmarking, kernel optimization, bottleneck analysis, low-level compute kernels

### LANGUAGES

Python, C++, C, Java, SQL

### AI HARDWARE

AI-accelerator software stacks, compute-kernel development, model tracing, sweep validation

### WEB & CLOUD

React, Vite, Tailwind CSS, Node.js, Cloudflare Workers

### TOOLING

Git & GitHub, CI/CD (GitHub Actions), pytest, PyPI, device/kernel profilers

### OTHER

Computer vision, machine learning, biometric systems, PCB design

## EDUCATION

### B.Tech, Electronics & Communication Engineering

Sri Manakula Vinayagar Engineering College, Puducherry  
2019 – 2023 • CGPA 8.5 / 10

## CERTIFICATIONS

- SkillRack — 300+ problem-solving medals in C & Java
- Python for Data Science — NPTEL
- Great Learning — C, Java, OS & Memory Management
- TCS iON — Career Edge

# ASWIN

## SOFTWARE ENGINEER

## PROFILE

Software engineer working close to the metal on the software that runs on next-generation AI accelerator hardware. I build and optimize compute kernels, profile and benchmark tensor operations, and hunt bottlenecks across the stack to drive device throughput. Equally comfortable owning end-to-end model tracing and validation and building the tooling, dashboards, and cloud infrastructure around the work.

## EXPERIENCE



### Software Engineer

Jun 2023 – Present

MulticoreWare Pvt Ltd • Chennai, India

- Software engineering for a leading AI silicon company's accelerator stack — developing and optimizing the software that runs on next-generation AI chips.
- Lead an element-wise operations subteam; drive kernel development, code review, and CI health for unary/binary element-wise ops.
- Author and optimize low-level compute kernels with broad data-type coverage (fp32, bf16, int32, uint8/16/32) and hardware-specific instruction-level work verified against ISA documentation.
- Built a model-trace sweep-validation system that reconstructs real model operation traces from a database and validates them against generated tests for exact-match coverage.
- Profile and benchmark kernels with the device profiler, tracking kernel execution time and diagnosing per-core load imbalance to guide optimization.
- Maintain performance-tracking automation and daily/weekly reporting pipelines that flag regressions against historical baselines and correlate them to commits.



### Industrial Project Engineer

Jun 2022 – May 2023

Lenovo Pvt Ltd • Pondicherry, India

- Developed and maintained a smart attendance system using face-recognition technology with database integration.
- Built an ESD tester with database-backed logging for real-time test workflows.
- Worked on computer-vision algorithms, biometric authentication, and real-time data processing.

## PROJECTS

### ttperf — AI-Accelerator Performance Profiler

CLI tool that profiles AI-accelerator compute tests and extracts device kernel performance metrics. Wraps the device profiler with pytest, parses result CSVs, and reports total kernel duration. Published on PyPI. **Python • pytest • PyPI** — [ttperf.aswincloud.com](https://ttperf.aswincloud.com)

### Eltwise Performance Tracker

Day-by-day performance dashboard for element-wise operations on AI-accelerator hardware — tracks trends across dates, correlates regressions to git commits, and sends automated alerts on >20% changes. **React • Vite • Tailwind • Recharts • Cloudflare Workers**

### Academic Projects — B.Tech ECE

Embedded & hardware builds: DDS-based portable function generator (up to 12 MHz), Arduino-based portable oscilloscope, 0–30 V / 0–2 A regulated power supply, and a non-contact AC/DC line tester.